Newer data like Sen1Floods11: where to get it

Checked on 18 Oct 2026. Every number below comes from a file we downloaded, an API we queried, or the dataset's own paper or page. Section 8 lists what we could not check.

What we wanted: data of the same type as Sen1Floods11, the 1.8 GB dataset used in `Flood_AI.ipynb`. That means Sentinel-1 radar and Sentinel-2 optical chips of a flood, each with a hand-drawn water map. We wanted it newer, with more of India, and still a manageable size.

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1. The answer in one minute

- **None of the newer public datasets we could check has more hand-labelled Indian chips than Sen1Floods11.** Sen1Floods11 still has the most: 68 chips from one flood (Assam, 2016).
- **Why:** most large global flood datasets take their labels from **Copernicus EMS** maps, Europe's emergency mapping service. This includes GEOID-Flood, STURM-Flood, ImpactMesh-Flood, MMFlood and WorldFloods. We listed all 1,067 Copernicus EMS activations. Since 2015 they include only **one Indian event**: Cyclone Fani, Odisha, May 2019. India maps its own floods through ISRO/NRSC, and those maps are not given out as open data.
- **So "updated data" for us comes from three places:**
    1. **C2S-MS Floods**, the closest thing to an updated Sen1Floods11.
        - It uses the same chip locations as Sen1Floods11 (445 of 446), with new hand labels made separately for radar and for optical, plus a cloud map.
        - It adds 7 more floods. Those 7 floods are **1.81 GB of GeoTIFFs**.
        - Free, no login.
    2. **Bigger, newer datasets for extra training.** None of them is built on an Indian flood, but some have nearby monsoon floods:
        - **GEOID-Flood (2026):** Pakistan 2022 and 2025, Sri Lanka 2017 and 2025. Nine of its Pakistan 2025 tiles cross the border into Indian Punjab, with 52 km² of checked flood labels.
        - **STURM-Flood:** 4 GB.
        - **SenForFlood:** has India, but its labels are mostly computer-made, and it is 289 GB.
    3. **Our own updated Indian set.** We tested every step.
        - Free radar and optical images come from Planetary Computer.
        - Free **Copernicus GFM** flood maps give the starting labels. GFM needs no login and covers all of India for every Sentinel-1 pass since 2015, including 2026.
        - We then hand-check a test subset.
        - **About 300 chips ≈ 1.8 GB, the same size as Sen1Floods11.** We already made the first 9 chips (Assam, June 2022).

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2. Everything we checked

All rows are also in `manifests/updated_datasets_checked_2026-10-18.csv`.

Dataset	Year	What is inside	Labels	India	Size	Licence	Fit for us
Sen1Floods11 (current)	2020	S1 + S2, 512 px, 446 chips, 11 floods (2016–19)	hand, all water	**68 chips** (Assam 2016) + 467 weak	1.77 GB	not stated, cite paper	base dataset
C2S-MS Floods	2022	S1 + S2, 512 px, **900 pairs, 18 floods** (2016–20)	**hand, per sensor**, + cloud	same 68 Assam chips, **new labels**	5.35 GB (3.47 GB GeoTIFF)	CC BY 4.0	**best update**
GEOID-Flood	2026	S1 GRD+RTC pre/post, S2 **pre-flood** composite, DEM; 1024 px; 14,282 tiles; 219 floods (2016–26)	manually validated: dry / normal water / flood	no Indian activation; 9 Pakistan-2025 tiles cover Indian Punjab (52 km² flood)	584 GB	CC BY 4.0	South Asian part
STURM-Flood	2025	S1 and S2 tiles, 128 px; 60 floods (Dec 2017–Oct 2022)	from Copernicus maps	0	3.99 GB	CC BY 4.0 / BY-SA	extra training
SenForFlood	2025, v2 Sep 2026	S1 + S2 before & during, DEM, land cover; 512 px; 40,770 samples; 350+ floods (2016–25)	**97% automatic**	yes (number not published)	289 GB	CC BY-SA 4.0	too big
ImpactMesh-Flood	2025–26	S1 RTC + S2 L2A time series, DEM; 24k samples, 206 floods (Dec 2015–Mar 2026)	from Copernicus maps	0 (has Pakistan 2022 + 2025, Sri Lanka 2025)	73 GB	CC BY 4.0	extra training
WorldFloods v2	2023	S2 + labels, 509 scenes	from Copernicus/UNOSAT maps	4 scenes (Fani 2019)	102 GB (India 0.12 GB)	**CC BY-NC**	already in our kit
Kuro Siwo	2023–24	**S1 only** (2 before, 1 after), 43 floods (2015–22)	manual	not listed	1,015 GB	MIT / CC BY	radar models only
MMFlood	2022	**S1 only** + DEM, 95 floods (2014–21)	Copernicus maps	0	11 GB	MIT	radar models only
S1S2-Water	2023	S1 + S2, 65 scenes	hand, all water bodies	0	174 GB	CC BY 4.0	water, not flood
FloodPlanet	2023–25	PlanetScope 3 m + S1, S2, Landsat; 366 tiles, 19 floods (2017–20)	hand	unknown	not stated	PDDL	not reachable from our server
Urban Flood Observations	2026	PlanetScope 3 m only, 215 chips, 14 floods	hand	0 (Dhaka 2020)	not checked	CC BY 4.0	different sensor
NASA / ETCI 2021	2021	**S1 only**	flood + water	0 (Bangladesh)	–	–	radar only
Sen1Floods11 + before-flood radar	2023	Sen1Floods11 chips + S1 before & after	Sen1Floods11	Assam (same)	2.2 GB	CC BY 4.0	change-detection idea
Copernicus GFM (a map service, not a dataset)	2015 → today	flood, water, normal water, "can't map here" mask, confidence; 20 m; every S1 pass	**automatic**	**all of India**	any box, any date	free, cite GFM	**labels for new Indian floods**
Microsoft AI4G floods	2014–24	S1 flood detections	automatic	all of India	~130 MB per tile	MIT	date finder

Dataset	Year	What is inside	Labels	India	Size	Licence	Fit for us
NRSC / NDEM / Bhuvan (ISRO)	ongoing	official Indian flood maps	expert	all of India	–	**view only**	checking by eye
India Flood Inventory (IIT Delhi)	1967–2023	table of flood events by district	records	all of India	1.9 MB	CC BY-NC	choosing events

S1 = Sentinel-1 radar. S2 = Sentinel-2 optical. "Pre-flood" means the image was taken before the flood, so it does not show the flood water.

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3. C2S-MS Floods: the updated Sen1Floods11

What it is. Cloud to Street and Microsoft published it in 2022, and it is hosted by Radiant Earth on AWS.

- 900 pairs of Sentinel-1 and Sentinel-2 chips, 512 × 512 px at 10 m.
- 18 floods:
  - **the 11 Sen1Floods11 floods** (445 pairs)
  - **7 new floods** (455 pairs): USA Texas (Hurricane Harvey) 2017, USA Nebraska 2019, Queensland 2019, Kenya/Uganda 2018, Czechia/Austria 2020, Niger 2019, Republic of the Congo 2020
- The full list is in `manifests/c2sms_floods_events.csv`.

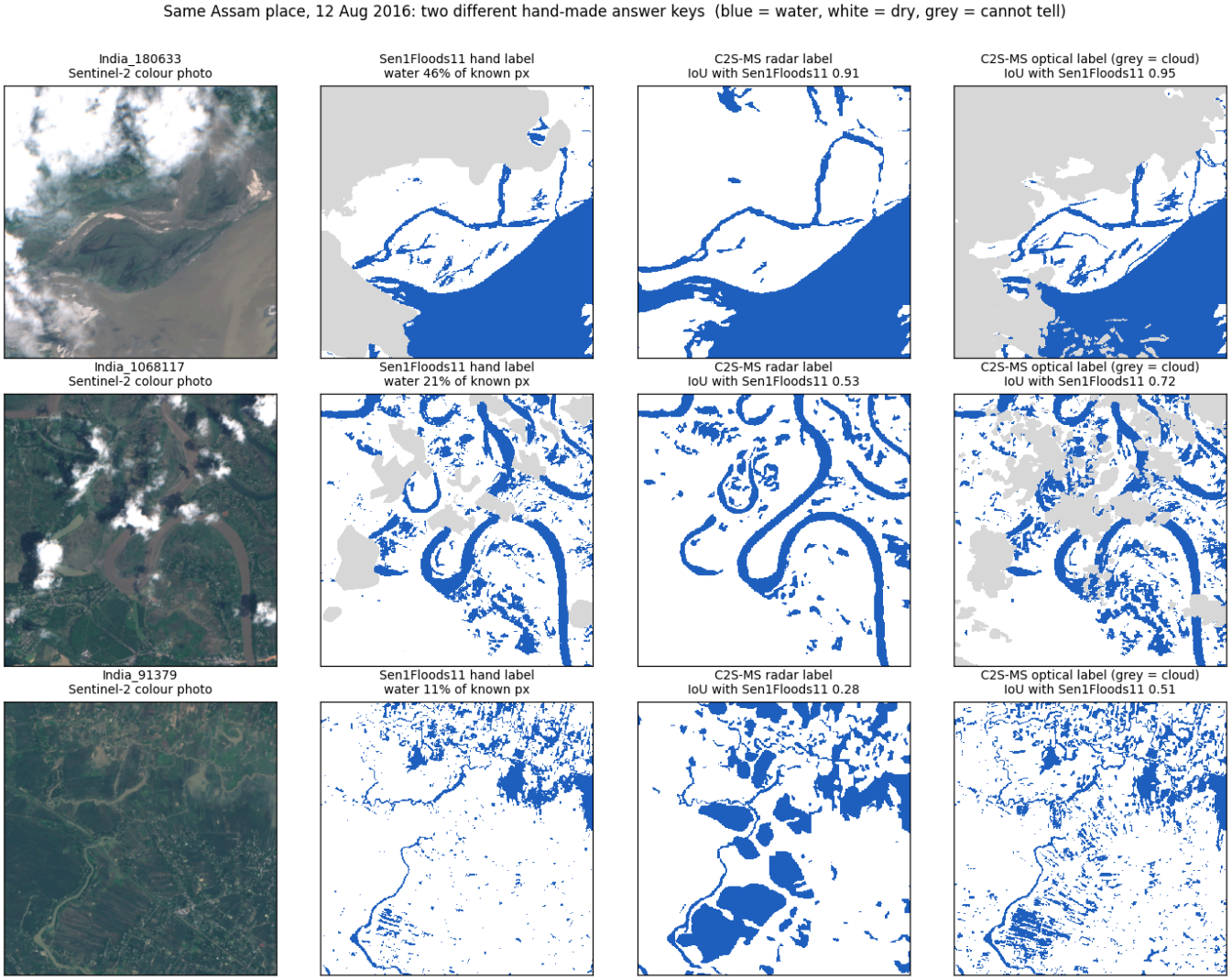
Why it is better than Sen1Floods11 as an answer key:

- **Two labels per place:**
  - `s1/LabelWater` shows the water a person could see in the radar image.
  - `s2/LabelWater` shows the water a person could see in the optical image.
  - Sen1Floods11 has one mixed label for both.
- **A cloud map:** `s2/LabelCloud` (0 clear, 1 cloud, 2 shadow).
- **Projection:** chips are in UTM metres (EPSG:326xx / 327xx), so a pixel is exactly 10 m.

What we checked on the 68 Indian chips (Assam, 12 Aug 2016):

Check	Result
Same places as Sen1Floods11?	**Yes.** Every one of the 68 Sen1Floods11 India chips lies 100% inside one C2S-MS chip, one-to-one. C2S-MS chips are a little larger (5.12 km UTM squares). For the other 10 floods the chip counts match (Paraguay has 66 instead of 67), but we did not check them chip by chip.
Same images?	Same dates: S1A 12 Aug 2016 23:46 UTC, S2A 12 Aug 2016 04:15 UTC.
Same labels?	**No, they were drawn again.** Agreement with the Sen1Floods11 hand label (IoU, all 68 chips pooled): **0.63 for the optical label** (cloud-free pixels), **0.52 for the radar label**. The C2S-MS radar and optical labels agree with each other at IoU 0.52.
Where they disagree	Water-rich chips agree very well. On the 7 India chips that are over 45% water, IoU is 0.92–0.97 for the optical label and 0.85–0.96 for the radar label. Most differences are in thin channels, small ponds and wet fields.
Cloud in the optical chips	24% of optical pixels are cloud or shadow on average.

Per-chip numbers are in `manifests/c2sms_vs_sen1floods11_india.csv`. The picture below shows three examples.



What this means for us:

1. C2S-MS India is **not a new Indian flood**. It is the same place and the same day, so do not count it as extra Indian test data.
2. It is a **second answer key**. Two careful human labels of the same Assam chips agree on about 92% of pixels, but their water areas overlap at only IoU 0.52–0.63. So when two models differ by a few points of IoU on India, that difference may be within the label noise. We should report scores against both answer keys.
3. The **7 new floods (1.81 GB of GeoTIFFs)** are extra hand-labelled training and validation data in a very similar format: 512 px, 10 m, S1 GRD dB + S2 L1C. The differences are a UTM grid, separate 0/1 radar and optical labels, and a separate cloud label instead of −1.

How to get it: no login, plain HTTPS.

```
        python3 scripts/09_download_c2sms_floods.py --events india --out data/c2sms   # 68 Assam
pairs, 0.26 GB
        python3 scripts/09_download_c2sms_floods.py --events new   --out data/c2sms   # 7 new floods,
1.81 GB
        python3 scripts/09_download_c2sms_floods.py --events all   --out data/c2sms   # all 18, 3.47
GB (GeoTIFFs only)
```

Processing note: S2 is L1C (top-of-atmosphere, like Sen1Floods11). S1 is GRD in dB.

---

4. Why the global datasets have almost no India

We downloaded the full list of Copernicus EMS mapping activations (1,067, rapid mapping plus risk & recovery) from the Copernicus API. The main South Asian flood and storm activations are listed below. All 19 are in `manifests/cems_south_asia_flood_activations.csv`.

Code	Date	Event	In GEOID-Flood	In STURM-Flood
EMSR040, 049, 054, 057, 104	2013–14	5 Indian events	no (before Sentinel-2)	no
EMSR357	**May 2019**	**Cyclone Fani, Odisha: the only Indian one since 2015**	no	no
EMSR205	May 2017	Sri Lanka floods	yes (23 tiles)	no
EMSR629, EMSR631	Aug–Sep 2022	Pakistan floods	yes (60 tiles)	yes (EMSR629)
EMSR838	Aug 2025	Pakistan floods	yes (228 tiles)	no
EMSR851	Nov–Dec 2025	Sri Lanka floods (Cyclone Ditwah)	yes (26 tiles)	no
EMSR164, EMSR439	2016, 2020	Bangladesh cyclone / flood	no	no
EMSR927	Aug 2026	Nepal flood	no (too new)	no

So a dataset built from Copernicus maps can contain only one Indian event, Cyclone Fani. Of the datasets we checked, only WorldFloods v2 has it (4 scenes, optical only).

There is one exception. The map areas of a neighbouring country's activation can cross the border. GEOID-Flood's Pakistan 2025 activation (EMSR838) covers part of Indian Punjab (see section 5).

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5. The bigger new datasets: what is worth taking

GEOID-Flood (2026): newest, manually checked

- **What is inside:**
  - 1024 × 1024 px tiles at 10 m
  - Sentinel-1 GRD and RTC, before and after the flood
  - one Sentinel-2 L2A **pre-flood** composite
  - Copernicus DEM
  - label: 0 dry, 1 normal water, 2 flood water, 255 not mapped
- **Splits** are by area of interest, and touching areas share a split. One activation can still have areas in both train and test (Pakistan 2025 has 15 test tiles), but the same ground is never in both.
- **Size and licence:** 584 GB, CC BY 4.0.
- **No Indian activation.** The South Asian part is 337 tiles:

Event	Country	Date	Tiles	Median flood share of mapped pixels
EMSR629 + 631	Pakistan	Aug–Sep 2022	60	51% / 11%
EMSR838	Pakistan	Aug 2025	228	6.5%
EMSR205	Sri Lanka	May 2017	23	1.3%
EMSR851	Sri Lanka	Nov–Dec 2025	26	2.2%

- **Labels only:** we downloaded the labels of the 337 South Asian tiles. 327 of them have a label file, and 319 are in the paper's selection. It was a 0.13 GB download, and about 5 MB was kept.
  - Per-tile list: `manifests/geoid_flood_south_asia_tiles.csv`
  - Command: `python3 scripts/10_geoid_south_asia.py --out data/geoid_south_asia`
- **Indian ground inside GEOID-Flood:**
  - We checked every South Asian tile against the Natural Earth country border.
  - **9 Pakistan 2025 tiles (EMSR838, images of 27–28 Aug 2025) have labelled pixels inside Indian Punjab.** They lie along the Sutlej near Fazilka and Ferozepur (73.9–74.6°E, 30.4–31.1°N).
  - Inside India they hold 128 km² of mapped ground, of which **52 km² is labelled flood**. All 9 are in the train split.
  - That is a small amount, but it is the only recent manually checked Indian flood label we found. The list is in `manifests/geoid_flood_tiles_inside_india.csv`.
- **With images:** `--layers label s1grd s2l2a dem`. About 40 GB of shards pass through, and about 9 GB of South Asian tiles is kept.
- **Limitation:** the optical image is from **before** the flood, so GEOID-Flood is mainly a **radar** benchmark. For radar + optical fusion, pull a during-flood S2 ourselves (script 04) on the same tile.

STURM-Flood (2025): small and easy

- 21,602 S1 tiles and 2,675 S2 tiles with water masks, from 60 Copernicus activations (Dec 2017–Oct 2022).
- One 3.99 GB zip on Zenodo.
- India: 0. Pakistan 2022 (EMSR629) is included: about 1,131 tiles, S1 and S2 together (2,262 files).
- Tiles are **128 px**, smaller than ours. Use it as extra training data after resizing or patching.

SenForFlood (2025, version 2 on 22 Sep 2026): has India, but labels are mostly computer-made

- 40,770 samples, 512 px at 10 m.
  - S1 and S2 (6 bands, L1C) both before and during the flood
  - DEM, slope, ESA WorldCover
- 350+ floods, 2016–2025. The paper names India, Bangladesh and Myanmar as a dense region.
- **39,431 masks are automatic.** They come from a Sentinel-1 before/after threshold on events listed by the Dartmouth Flood Observatory (DFO). Only 1,339 masks come from Copernicus maps.
- **289 GB in total.** The original archive has 114 parts (272 GB, April 2025), and you need all of them. Version 2 added 8 "v1.1" files (17 GB) on 22 Sep 2026, and their contents are not described. Licence is CC BY-SA 4.0.
- Worth it later if we want many automatic Indian samples. Not needed now.

Not useful for our exact task

- **Kuro Siwo** and **MMFlood:** radar only.
- **S1S2-Water:** all water bodies, not floods.
- **FloodPlanet** and **Urban Flood Observations:** PlanetScope 3 m.

- **ImpactMesh-Flood:** good, but 0 India. It does have Pakistan 2022 (296 samples), Pakistan 2025 (414) and Sri Lanka 2025 (33).

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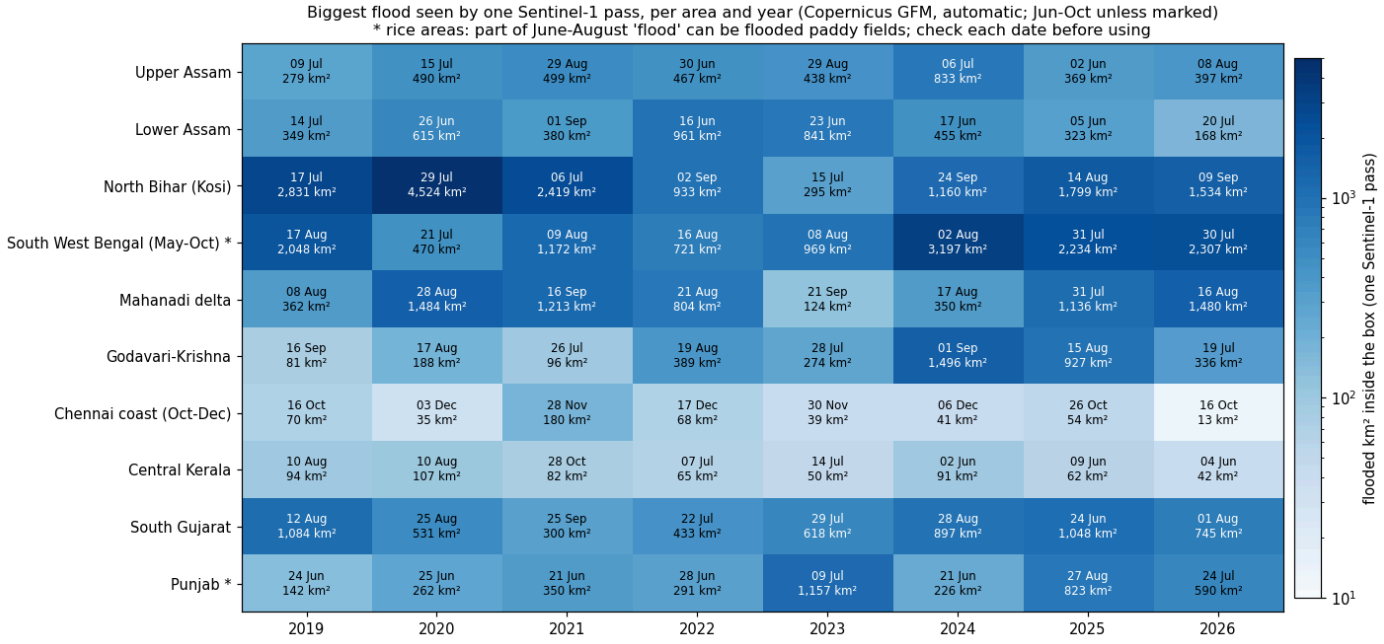
6. Build our own updated Indian set (recommended)

Steps 1–5 were run here and worked. Step 6 is manual work.

Step 1: choose recent Indian floods (script 07)

`scripts/07_find_india_floods_gfm.py` looks through the Copernicus GFM archive for 10 flood-prone Indian areas, for every year 2019–2026. GFM is a set of automatic flood maps made from every Sentinel-1 pass since 2015; we searched it through the public EODC catalogue. For each satellite pass, the script measures how many km² GFM marked as flooded inside each area. The full list is in `manifests/india_flood_candidates_gfm.csv`.

Result: 3,673 satellite passes over the 10 areas, 2019–2026. The picture shows the biggest pass in each area and year.



Many peaks match well-known floods. These are the ones we suggest for the new Indian set:

Area	GFM peak pass	Flooded km² in the box	Coincides with (to check)
Lower Assam	16 Jun 2022	961	June 2022 Assam floods (Sentinel Asia activation on 15 Jun 2022)
Upper Assam	6 Jul 2024	833	July 2024 Assam floods
North Bihar (Kosi)	29 Jul 2020	4,524	July 2020 Bihar floods
North Bihar (Kosi)	24 Sep 2024	1,160	Kosi floods, Sep–Oct 2024
Mahanadi delta	28 Aug 2020	1,484	Aug 2020 Odisha floods
Mahanadi delta	16 Sep 2021	1,213	Sep 2021 Odisha floods (Microsoft AI4G also found 14–16 Sep 2021)

Area	GFM peak pass	Flooded km² in the box	Coincides with (to check)
Godavari–Krishna	1 Sep 2024	1,496	Vijayawada (Budameru) floods, Sep 2024
South Gujarat	28 Aug 2024	897	Gujarat floods around Vadodara, late Aug 2024
Punjab	9 Jul 2023	1,157	Punjab floods, July 2023
Punjab	27 Aug 2025	823	Punjab floods, Aug–Sep 2025
2026 (newest)	Bihar 9 Sep, Mahanadi 16 Aug, Gujarat 1 Aug 2026	1,534 / 1,480 / 745	not yet checked against reports

We matched these dates to widely reported floods. Check each one on the NRSC maps (Bhuvan / NDEM) or in news reports before final use.

Things to watch:

- In rice areas, flooded paddy fields in June–August look like flood water to radar. This affects South West Bengal and Punjab in particular. The June peaks in Punjab in 2019–2022 and 2024 are probably paddy, not floods.
- Cities are inside GFM's "can't map here" mask, so urban floods (for example Chennai) barely show.
- Kerala's narrow valleys give small numbers. Its big 2018 flood is before 2019.

GFM numbers are automatic and were measured on a coarse 640 m view. Use them to pick dates, then look at the images.

Step 2: pick small boxes with the most flood (script 12)

The areas in step 1 are large, about 100–300 km across. Script 12 cuts an area into 0.18° squares and lists the squares GFM saw as most flooded on that date. A square is about 18 × 20 km, which fits 3 × 3 chips up to 32°N. It prints ready-to-run commands for script 04.

```
        python3 scripts/12_gfm_flood_boxes.py --date 2022-06-16 --bbox 89.8 25.8 92.2 26.8 --top 4 --
name India_LowerAssam --out boxes.csv
```

For Lower Assam on 16 Jun 2022 the four best squares were around Barpeta (91.06–91.60°E, 26.26–26.62°N), with 24–33% flooded.

Step 3: download the images (script 04, already in the repo)

- Sentinel-1 RTC, Sentinel-2 L2A and JRC normal water, from Microsoft Planetary Computer. No login.
- In the monsoon, Sentinel-2 is usually cloudy. For the Mahanadi flood of 16 Aug 2022, the nearest Sentinel-2 image (15 Aug) was 97% cloud. In Assam (15 Jun 2022) it was 100%. Expect radar-only chips for most dates.

Step 4: add starting labels from Copernicus GFM (script 08)

```
python3 scripts/08_pull_gfm_labels.py --pull data/india_new/<event>
```

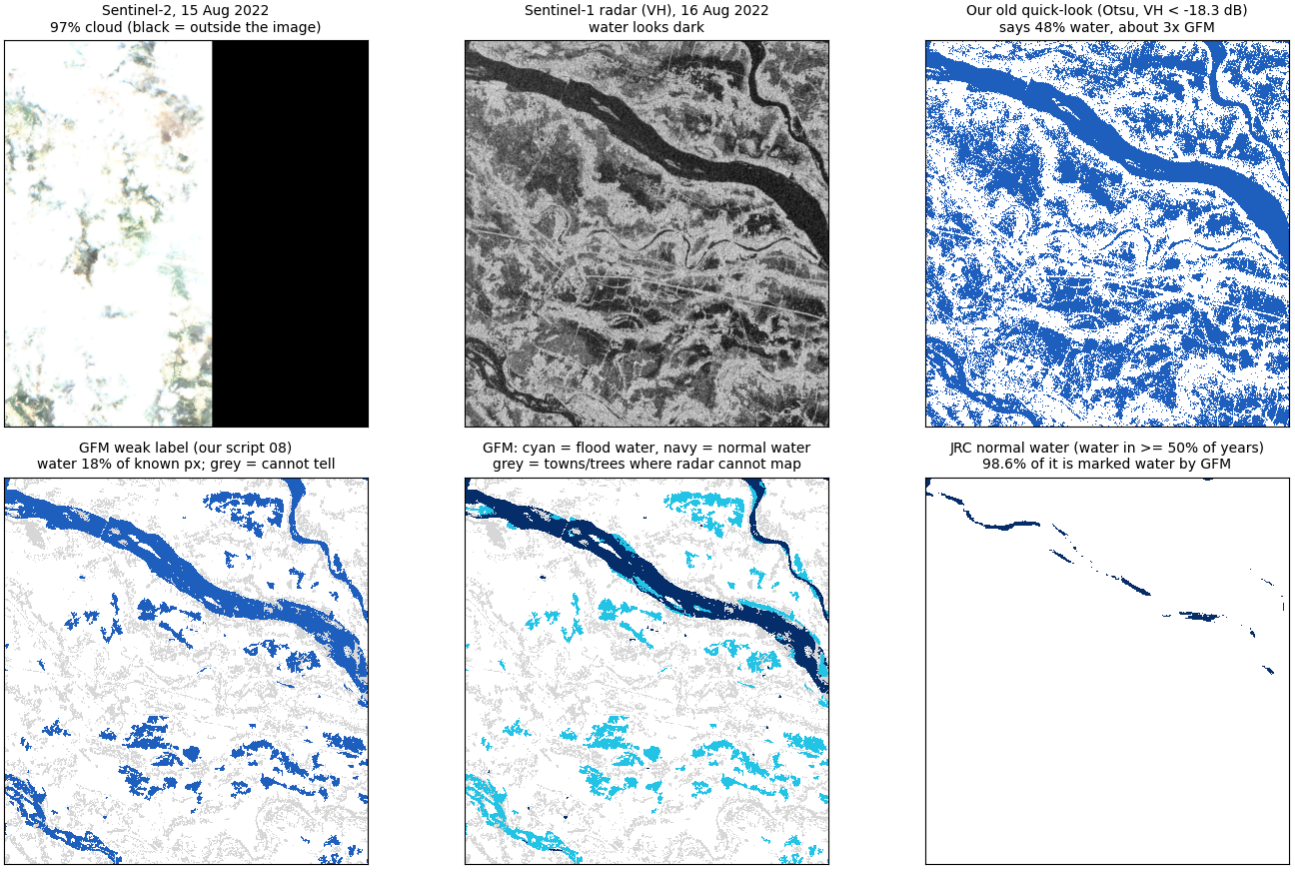
- It looks for the GFM map of the **same Sentinel-1 pass** and puts these layers on our 10 m grid:
  - flood water
  - all water

    - normal water
    - an "exclusion" mask (towns, dense trees and radar shadow, where radar cannot see water)
    - a confidence layer
- If it can't find that pass, it uses another pass from the same date. `meta.json` records which one it used (`same_scene_as_radar`).
- It also writes `label_gfm.tif` in the Sen1Floods11 style: 1 water, 0 dry, −1 can't tell.

Test on the Mahanadi flood of 16 Aug 2022:

	Result
GFM map found for the same radar pass as our image	yes (S1A 2022-08-16 00:05 UTC)
Water	17.6% of known pixels = about 10.1% flood + 7.5% normal water (flood is 8.4% of the whole box)
Can't tell (exclusion mask)	16.8% of the box
JRC normal-water pixels that GFM marks as water	98.6%
Our old radar quick-look (Otsu threshold)	47.6% water, about 3× more than GFM

Mahanadi delta near Cuttack, flood of 16 Aug 2022: the free Copernicus GFM map gives cleaner starting labels than our quick-look

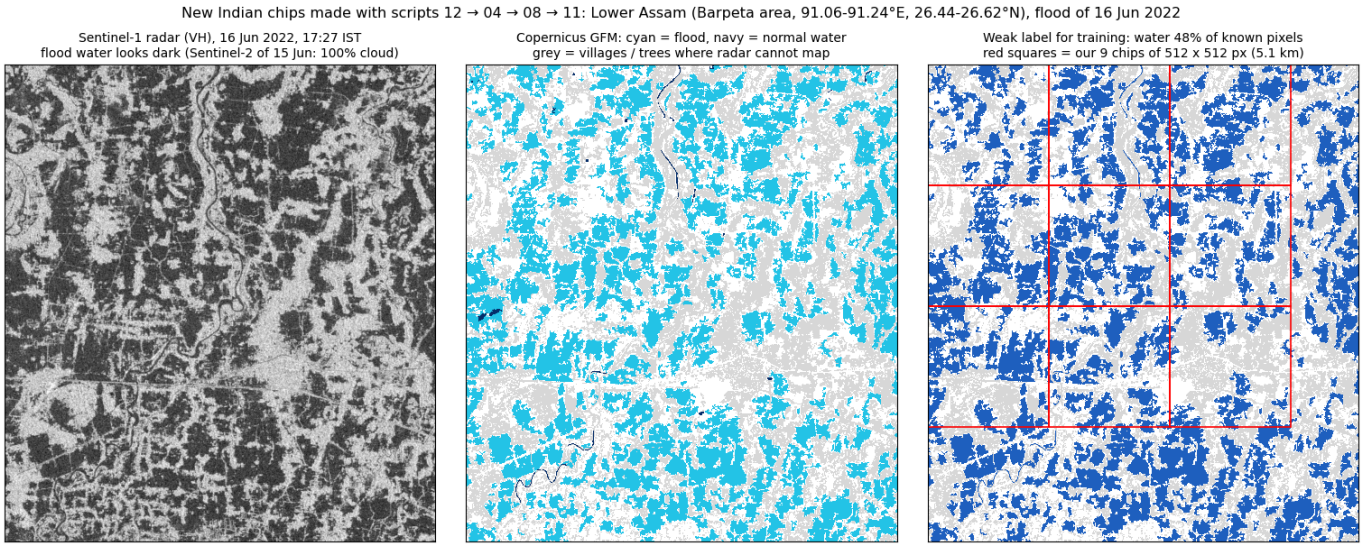


GFM labels are much cleaner than a simple threshold. They are still computer-made at 20 m, so they are **training labels only**.

Step 5: cut chips (script 11)

```
python3 scripts/11_make_chips.py --pull data/india_new/<event> --event India_<Place>_<YYYYMMDD> --out data/india_chips
```

- It writes Sen1Floods11-style folders: `S1RTC/`, `S2L2A/`, `LabelGFM/`, `JRCWater/`, `S2SCL/`, plus `chips.csv`.
- **Measured size:** 5.0 MB per chip at the Mahanadi and 6.2 MB per chip in Assam. Allow about 6 MB per chip, so **300 chips ≈ 1.8 GB**.
- **First real test: Lower Assam, 16 Jun 2022.**
  - Box 1 from step 2 (91.06–91.24°E, 26.44–26.62°N) gave **9 chips**.
  - Sentinel-2 was 100% cloud and radar was 100% valid.
  - GFM marks 48% of known pixels as water, and 31% of the box is "can't tell" (villages with trees).



The whole recipe for one flood:

```
        python3 scripts/07_find_india_floods_gfm.py --years 2019 2026 --out
manifests/india_flood_candidates_gfm.csv   # once: find dates
        python3 scripts/12_gfm_flood_boxes.py --date 2022-06-16 --bbox 89.8 25.8 92.2 26.8 --top 4 --
name India_LowerAssam --out boxes.csv
        python3 scripts/04_pull_india_s1s2.py --bbox 91.06 26.44 91.24 26.62 --date 2022-06-16 --days
1 --out data/india_new/India_LowerAssam_20220616_box1
        python3 scripts/08_pull_gfm_labels.py --pull data/india_new/India_LowerAssam_20220616_box1
        python3 scripts/11_make_chips.py --pull data/india_new/India_LowerAssam_20220616_box1 --event
India_LowerAssam_20220616_box1 --out data/india_chips
```

Step 6: hand-check a test set

- Choose about **80–100 chips from 3 floods** that are **never used for training**. Correct the GFM label by hand.
- Help for checking:
  - the Sentinel-2 image when it is clear
  - NRSC flood maps on Bhuvan / NDEM (view only)
  - Sentinel Asia products for some Indian floods. For example, Assam June 2022 has EOS-RS flood proxy maps (automatic, KMZ/SHP).
- Save the corrected labels as `LabelHand/`, in the same format as Sen1Floods11.

---

7. The plan: a new "India pack" of about 1.8 GB, on top of Sen1Floods11

Part	What	Size	Use
A	Sen1Floods11, unchanged (same as `Flood_AI.ipynb`)	1.77 GB	base for all 5 models
B	C2S-MS India labels (second answer key for the 68 Assam chips)	about 1.3 MB of labels (0.26 GB with images)	fairer India score
C	**New Indian chips 2019–2026**: about 10 floods × 30 chips ≈ 300 chips (S1 RTC + S2 L2A + GFM label + JRC + scene classes)	**≈ 1.8 GB**	training (GFM labels)
D	**Hand-checked Indian test set** from 3 of those floods (80–100 chips)	inside C	the honest "new Indian flood" test
Optional	C2S-MS 7 new floods (1.81 GB), STURM-Flood (3.99 GB), GEOID South Asia including the Indian Punjab tiles (labels 5 MB, images ~9 GB)	2–15 GB	more training if time allows

Suggested floods for part C are the 10 dated floods in the table in step 1. Add one or two 2026 floods once they are checked.

Use **3 floods in different states as the hand-checked test set**, for example Assam 2022, Vijayawada 2024 and Gujarat 2024. Train on the rest.

Rules:

1. Split by **flood**, never by chip.
2. Never score models on GFM labels.
3. Sen1Floods11 and C2S-MS use S2 L1C + S1 GRD dB. New pulls use S2 L2A + S1 RTC dB. Normalise each source separately.
4. Monsoon chips are cloudy, so every model must also work with radar only.
5. Licences:
    - WorldFloods v2 and the India Flood Inventory are non-commercial.
    - SenForFlood and the STURM-Flood repository are share-alike.
    - Cite GFM, Copernicus and each dataset.

---

8. What we could not check

- **FloodPlanet:** the CyVerse server redirected our cloud machine to a block page. Try downloading from the IIT Bombay network. The list of its 19 flood locations is not public in the pages we could read.
- **SenForFlood:** the paper confirms India but does not give a count. Counting needs the full 272 GB original archive.
- **SenForFlood numbers:** the sample counts (40,770 / 39,431 / 1,339) and the band list come from the paper text as we read it. We could not check them in the files.
- **Kuro Siwo:** the full event list is only in a GitHub folder we were not allowed to read. The paper names Sri Lanka and Nepal events but not India.
- **Sen2GF3Floods** (*Scientific Data*, 2026; Sentinel-2 + Chinese GF-3 radar): the article page did not load.
- **NRSC/NDEM GIS layers:** the public portals show the maps but do not offer downloads. NRSC sends "GIS-ready" layers to state agencies (for Assam, to ARSAC and NESAC, as stated in an ASDMA flood report). An official request through IIT Bombay may get them.

- **GFM licence:** the EODC catalogue lists the licence as "proprietary". GFM is part of Copernicus EMS and is free to use with credit. Check the GFM terms before using it in any ClimMaTech product.

Sources

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